

National University of Computer and Emerging Sciences

CS 3009 – Software Engineering

Project Deliverable 0

Delivered To: Ms. Ansa Liaqat

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PharmaX

Pharmaceutical Distribution Management System | Desktop Application

1. Problem Statement

Pharmaceutical distributors play a critical role in ensuring the timely availability of medicines to pharmacies, clinics, and hospitals across Pakistan. Despite operating in a compliance-heavy, high-stakes environment regulated by DRAP (Drug Regulatory Authority of Pakistan) and FBR (Federal Board of Revenue), many mid-to-large distributors still rely on fragmented tools — spreadsheets, paper-based registers, and disconnected billing software — to manage their day-to-day operations.

This fragmented approach leads to a range of serious operational problems. Without real-time stock visibility, distributors frequently face stockouts or overstock situations, both of which directly impact profitability and service reliability. Batch-level tracking is often manual or absent, making it extremely difficult to identify and act on near-expiry medicines before financial losses occur.

Billing is another major pain point. Generating FBR-compliant tax invoices manually — while correctly applying batch-specific rates, trade schemes, and discounts — is time-consuming and error-prone. This creates compliance risks and disputes with both customers and suppliers. Customer credit management is equally problematic: without automated credit limit tracking and AR ageing analysis, overdue receivables go unnoticed and cash flow suffers.

On the supplier side, distributors struggle to reconcile GRN records against payments made, leading to duplicate payments, missed credit notes, and strained supplier relationships. Additionally, without a centralised audit trail, any DRAP inspection or FBR tax review becomes a labour-intensive exercise. The cumulative effect of these gaps is reduced profitability, heightened compliance exposure, and an unreliable supply chain that ultimately affects patient access to medicine across Pakistan.

2. Problem Solution and Scope

PharmaX is a desktop-based, centralised application built specifically for Pakistani pharmaceutical wholesale distributors. It addresses the core operational gaps by providing an integrated platform for master data management, purchase and sales processing, real-time inventory tracking, financial ledgering, and data export — all within a single, role-based application.

The system delivers real-time visibility into stock levels and batch details (including expiry dates and DRAP drug schedules), automated FBR-compliant billing with support for trade schemes and discounts, and strong credit control through customer-wise limits and AR ageing analysis. Supplier and customer ledgers are maintained with full accuracy, and a complete audit trail covers all GRNs, sales, returns, and payments.

In scope: product, supplier, customer, and batch master data; purchase/GRN management; sales invoicing with FEFO batch selection; stock and expiry tracking; financial ledgers (receivables and payables); basic expense recording; and data export to Excel/CSV for reporting and DRAP/FBR audit. Out of scope: mobile apps, web-based customer portals, manufacturing or production features, direct pharmacy POS integration, and advanced predictive analytics or AI forecasting. Target users are medium to large pharmaceutical wholesale and distribution businesses operating in Pakistan.

3. Core Features

The following features form the functional core of PharmaX (excluding login and sign-up):

#	Feature Name	Description
1	Add / Manage Products	Create and update medicine master records: name, category, manufacturer, pack size, HSN code, DRAP drug schedule, FBR GST rate, MRP (batch-level), and default pricing rates.
2	Add / Manage Batches	Record incoming batches with batch number, manufacturing date, expiry date, MRP, purchase cost, quantity, and supplier/GRN link. Opening stock movement auto-created.
3	Add / Manage Suppliers	Maintain supplier profiles including name, STRN (Sales Tax Registration Number), NTN (National Tax Number), DRAP wholesale drug licence, address, city, payment terms, and credit period.
4	Add / Manage Customers	Store customer details such as name, STRN, NTN, DRAP drug licence, credit limit, payment terms, territory, city, and customer type (retailer / wholesaler / hospital / clinic / government).
5	Record Sales Invoice	Create FBR-compliant sales bills with FEFO batch selection, automatic quantity/rate calculation, scheme/discount support, GST computation, and real-time stock deduction.
6	Expiry Management	Auto-tracks all batch expiry dates; shows configurable alerts (30/60/90 days before expiry); prevents sale of expired stock. Critical/Warning/Watch status classification.
7	Customer Ledger	Maintains a running account per customer covering sales, returns, payments received, current balance, AR ageing analysis (0–30 / 31–60 / 61–90 / 90+ days), and credit limit status.
8	Supplier Ledger & Payments	Tracks GRNs, purchase returns, and payments made to suppliers, showing outstanding bills, payment history, and accounts payable (AP) status.

9	Expense Tracker	Records non-inventory operational expenses (transport, rent, utilities, salaries, etc.) with date, amount, category, and payment mode (Cash / Bank Transfer / Cheque / JazzCash / EasyPaisa).
10	Data Export	Exports master data, transactions, ledgers, and stock summaries to Excel/CSV format for FBR audit, DRAP compliance, backup, and external reporting.

4. Project Plan

a. Modular Breakdown — 3 Iterations (2 Weeks Each)

Iter	Weeks	Module	Main Deliverable	Key Features	User Stories
1	1–2	Authentication & Master Data	Working desktop app with secure login and complete master data setup	Login/Sign-Up, Products, Suppliers, Customers, Batches, Basic Stock View	US-101–US-106
2	3–4	Purchase & Sales Core	Full transactional engine with stock and ledger updates	Purchase GRN, Sales Invoice, Returns, Expiry Alerts, Customer & Supplier Ledgers	US-201–US-207
3	5–6	Financials, Reports & Export	Complete financial closure: expense tracking, payments, dashboards, and export	Expense Entry, Payments/Receipts, Financial Reports, Dashboard, Data Export	US-301–US-306

Dependencies between iterations:

- Iteration 2 requires Iteration 1: Master data (products, suppliers, customers, batches) must exist before any GRN or sales invoice can be created or processed.
- Iteration 3 requires Iteration 2: Transactions and ledger entries from Iteration 2 feed into the financial reports, analytics dashboards, and export functionality of Iteration 3.

User Stories

Iteration 1 — Foundation & Master Data Management

- As a distributor, I want to register and log in to the system so that I can access PharmaX securely with my role-based permissions.
- As an admin, I want to add new medicines to the product master (with DRAP schedule, HSN code, and FBR GST rate) so that the system has accurate and complete product information.
- As an admin, I want to register suppliers with complete details (STRN, NTN, DRAP drug licence) so that I can maintain reliable supplier records for purchasing.
- As an admin, I want to register customers with credit terms and territory so that I can manage customer relationships and credit policies.
- As a distributor, I want to add new batches (with MRP, manufacturing and expiry dates) so that I can track stock accurately by batch in compliance with DRAP requirements.
- As a distributor, I want to view current stock levels across all products so that I know exactly what inventory is available at any time.

Iteration 2 — Purchase & Sales Operations

- As a distributor, I want to create GRNs (Goods Receipt Notes) from suppliers and record batch details so that I can accurately record incoming stock and costs.
- As a distributor, I want to generate FBR-compliant sales invoices to customers with automatic GST calculation so that I can bill customers correctly and compliantly.
- As a distributor, I want to record purchase returns and update inventory accordingly so that stock and financial records remain accurate after returns.
- As a distributor, I want to process sales returns and adjust customer credits so that customer accounts and inventory reflect returned goods properly.
- As a distributor, I want to receive alerts for near-expiry batches (30/60/90 days) so that I can take action to sell or return expiring stock in time.
- As a distributor, I want to view customer payment history and outstanding dues (AR ageing) so that I can manage credit control and follow up on payments effectively.
- As a distributor, I want to view supplier payment history and outstanding liabilities (AP) so that I can manage cash flow and supplier relationships responsibly.

Iteration 3 — Financial & Analytics Module

- As a manager, I want to record all operational expenses and categorise them so that I can track and control business running costs.
- As a finance user, I want to record payments made to suppliers (via cash, bank transfer, cheque, JazzCash, or EasyPaisa) and track payment status so that supplier accounts stay up-to-date.
- As a finance user, I want to record customer payments and update their outstanding balance so that customer ledgers remain accurate and current.
- As a manager, I want to generate profit and loss statements for a given period so that I can understand the financial performance of the business.
- As a manager, I want to export all data to Excel format so that I can perform detailed DRAP/FBR auditing, further analysis, and external reporting.
- As a manager, I want to view a dashboard with key performance indicators and trends so that I can quickly assess business health and make informed decisions.

b. Team Roles & Responsibilities

Role	Team Members	Primary Responsibilities	
Project Lead & Architect	M Abdullah	Project direction, scope control, system architecture, database design, final integration	
Backend Developer (Senior)	Ali Hassan & Sheraz Malik	Business logic, database layer, purchase/sales/ledger modules (mainly Iteration 2)	
Frontend / UI Developer	M Abdullah & Ali Hassan	Desktop UI screens, layouts, usability, visual consistency across all modules	
Database Admin & Tester	Ali Hassan, M Abdullah & Sheraz Malik	Database schema, data integrity, unit + integration testing, bug tracking	
Requirement Engineer & Documenter	M Abdullah & Sheraz Malik	User stories, feature specifications, validation, documentation and user guide	

